

***O+O and d+Au centrality using different
initial model inputs or multiplicities from
mid and forward rapidities with $\langle N_{part} \rangle$
 $\langle N_{coll} \rangle$ systematics study***

Zhengxi Yan

Centrality definition meeting (Sep 9, 2025)

Initial models for nucleon position

Previous study on the raw distribution of N_{ch} (TPC charged tracks within $|\eta| < 1.5$, and total MIP within EPD. Event and track level cuts are determined.

<https://drupal.star.bnl.gov/STAR/blog/zyan/dAu-2021-centrality-study>

<https://drupal.star.bnl.gov/STAR/blog/zyan/Centrality-study-OO>

How do we model initial nucleon positions in Glauber simulations?

For O+O, we have 2 ab initio models EFT and VMC, and best-fit 3 parameter Fermi distribution (3PF): 3PFVMC and 3PFEFT

For d+Au, we simply use the default Glauber configuration for d and Au. Here d use Hulthén function and Au use 3PF function with default parameters.

Todos:

1. Take each of those as input to compared to data
2. Do NBD sampling and find the best parameters that describes each
3. Compare the centrality cut difference
4. Further study systematics for $\langle N_{part} \rangle$ and $\langle N_{coll} \rangle$

Note:

Previous works(from above 2 links) only studied the centrality cuts but not $\langle N_{part} \rangle$ and $\langle N_{coll} \rangle$. Due to improper backup of the codebase, the exact method to match those results were not recovered. All new NBD fit were done from scratch and the best fit is found only considering the data. Please use the new centrality cuts for guarantee on reproducibility. Old fit, denoted as Old Sim, are also shown along the new fits for information purposes.

O+O

O+O Nch cuts

Percentile(%)	Old cuts(3PFEFT)	EFT	VMC	3PFEFT	3PFVMC
0 - 1	121.20	122.64	122.43	121.80	122.16
1 - 2	111.19	112.76	112.53	111.81	112.22
2 - 3	104.61	106.30	106.04	105.25	105.70
3 - 4	99.52	101.32	101.04	100.19	100.68
4 - 5	95.29	97.05	97.00	95.71	96.67
5 - 6	91.62	93.52	93.43	92.15	93.10
6 - 7	88.34	90.37	90.27	88.96	89.89
7 - 8	85.36	87.53	87.38	86.06	86.97
8 - 9	82.61	84.90	84.75	83.38	84.29
9 - 10	80.04	82.45	82.27	80.87	81.80
10 - 15	69.13	72.06	71.71	70.22	71.14
15 - 20	60.25	63.46	63.07	61.45	62.40
20 - 25	52.70	56.08	55.57	53.87	54.87
25 - 30	46.19	49.56	48.95	47.26	48.23
30 - 35	40.58	43.75	43.05	41.42	42.33
35 - 40	35.73	38.55	37.76	36.26	37.06
40 - 45	31.55	33.87	33.04	31.67	32.39
45 - 50	27.94	29.67	28.82	27.62	28.23
50 - 55	24.81	25.89	25.01	24.02	24.49
55 - 60	22.05	22.49	21.57	20.78	21.16
60 - 65	19.59	19.39	18.45	17.87	18.15
65 - 70	17.34	16.54	15.59	15.20	15.39
70 - 75	15.21	13.89	12.94	12.73	12.82
75 - 80	13.15	11.39	10.45	10.40	10.41
80 - 85	11.07	8.99	8.06	8.16	8.11
85 - 90	8.89	6.59	5.70	5.93	5.82
90 - 95	6.42	4.03	3.23	3.55	3.39
95 - 100	0.00	0.00	0.00	0.00	0.00

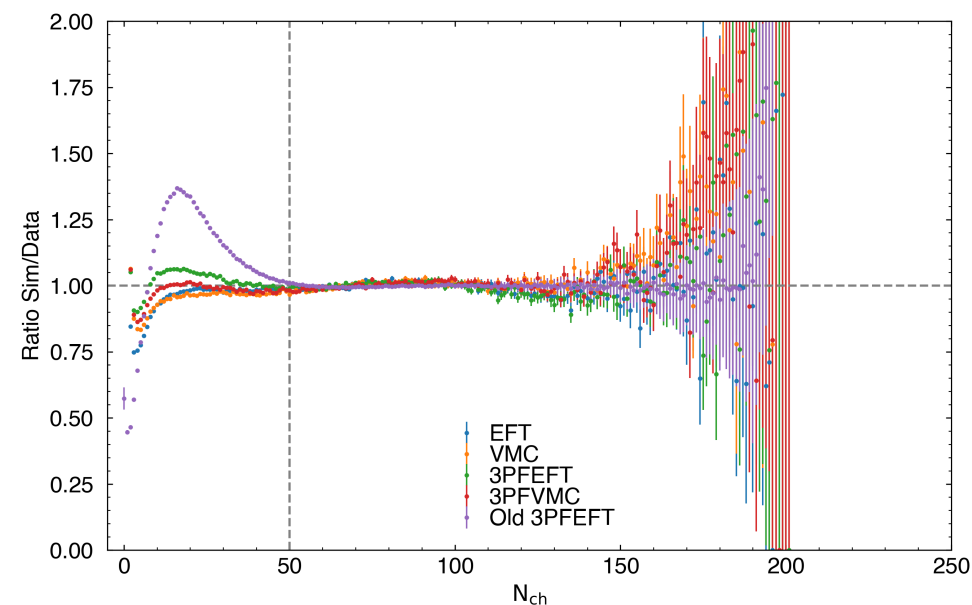
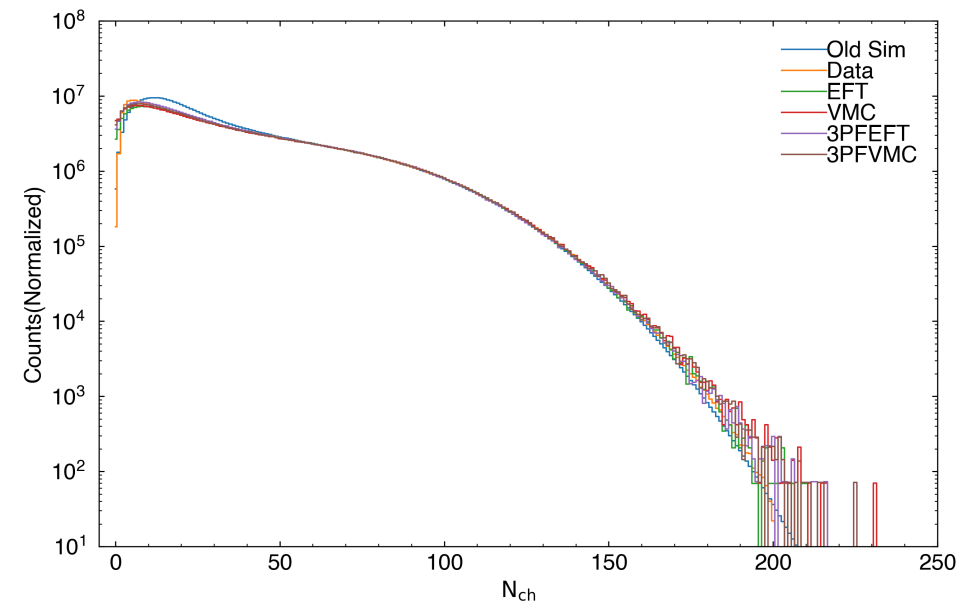
Glauber Fit Parameters:

EFT: npp=6.8, k=3.0, x=0.14, eff=0.14

VMC: npp=6.4, k=2.0, x=0.14, eff=0.14

3PFEFT: npp=6.9, k=2.67, x=0.14, eff=0.14

3PFVMC: npp=6.6, k=2.33, x=0.14, eff=0.14



Fit range > 50

O+O Total MIP cuts

Percentile(%)	Old	cuts(3PFEFT)	EFT	VMC	3PFEFT	3PFVMC
0 - 1	449.08	453.29	452.06	451.05	451.72	
1 - 2	418.30	423.24	421.81	420.64	421.41	
2 - 3	397.51	403.22	401.62	400.31	401.18	
3 - 4	381.12	387.52	385.77	384.33	385.28	
4 - 5	367.25	374.39	372.77	371.14	372.55	
5 - 6	355.03	362.64	360.70	359.13	360.60	
6 - 7	343.97	352.00	349.83	348.30	349.87	
7 - 8	333.79	342.29	339.72	338.31	339.91	
8 - 9	324.29	333.21	330.42	329.04	330.76	
9 - 10	315.34	324.64	321.71	320.34	322.12	
10 - 15	276.08	287.29	283.71	281.78	283.83	
15 - 20	242.64	255.60	251.58	248.97	251.34	
20 - 25	213.03	227.59	223.23	220.05	222.65	
25 - 30	186.53	202.31	197.85	194.12	196.90	
30 - 35	162.79	179.55	174.79	170.86	173.65	
35 - 40	141.57	158.88	153.97	150.15	152.72	
40 - 45	122.68	140.15	135.18	131.72	133.91	
45 - 50	105.92	123.16	118.24	115.19	117.13	
50 - 55	91.09	107.84	102.96	100.51	102.03	
55 - 60	77.96	93.93	89.24	87.41	88.54	
60 - 65	66.29	81.36	76.67	75.55	76.33	
65 - 70	55.84	69.76	64.93	64.62	65.05	
70 - 75	46.36	58.94	53.83	54.44	54.51	
75 - 80	37.61	48.72	43.24	44.78	44.53	
80 - 85	29.32	38.81	33.01	35.42	34.90	
85 - 90	21.19	28.93	22.90	26.15	25.36	
90 - 95	12.73	18.41	12.54	16.38	15.34	
95 - 100	0.00	0.00	0.00	0.00	0.00	

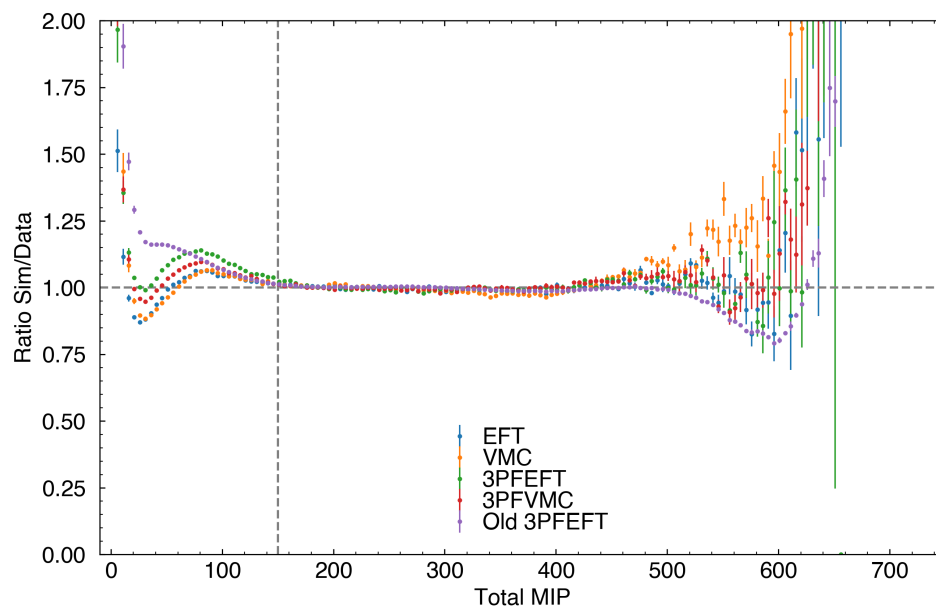
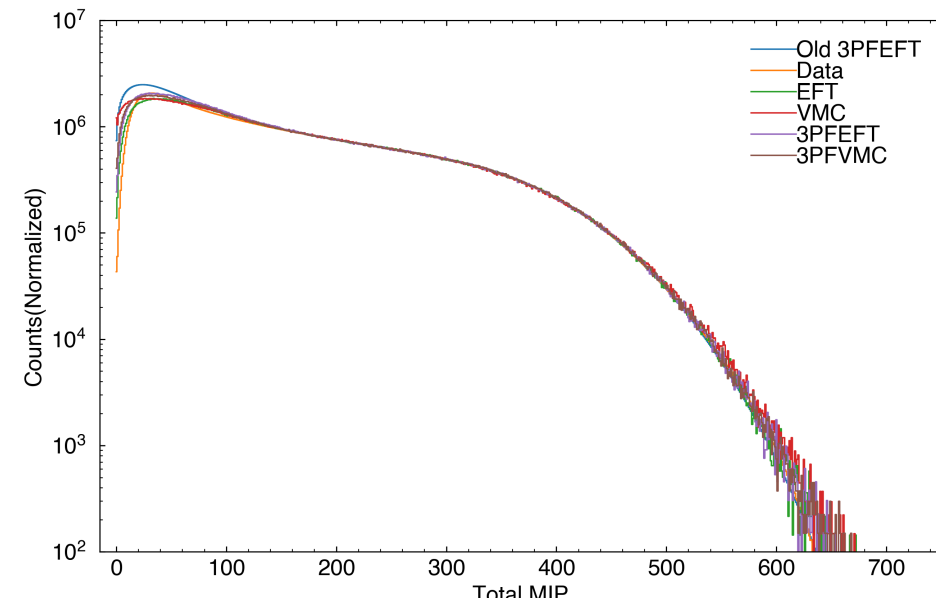
Glauber Fit Parameters:

EFT: npp=29.67, k=2.67, x=0.14, eff=0.14

VMC: npp=29.44, k=1.33, x=0.14, eff=0.14

3PFEFT: npp=30.56, k=2.33, x=0.14, eff=0.14

3PFVMC: npp=29.22, k=2.0, x=0.14, eff=0.14



Fit range > 150

Systematic error on <Npart> and <Ncoll>

Systematic variations

1. Vary nuclear shape (Woods-Saxon radius R and skin depth d in opposite direction. Large = R up and d down. Small = R down and d up.
 - a. In O+O, for 3PF models only. In d+Au, for Au nucleus only.
2. Vary the inelastic nucleon-nucleon cross-section by +/- 1 mb. Default 42 mb.
3. Assume a Gaussian nuclear profile instead of Woods-Saxon for wounded nucleons determination.
4. Vary the NBD particle production parameter Npp by +/- 5%.
5. Vary the total interaction cross-section by +/- 5%.

Total Cross-Section Variation Method (as in the standard code)

1. Centrality is defined by slicing the total multiplicity distribution into percentile bins.
2. A +/- 5% cross-section variation is implemented by shifting the multiplicity values that define these percentile boundaries.
3. This effectively changes the multiplicity selection for each centrality bin.
(*This source cause <Npart> <Ncoll> change that is larger in 11.5 GeV Au+Au compared to 200 GeV O+O and d+Au studied here)

Total Systematic Error Calculation

$$\Delta_{sys} = \sqrt{\sum_i (O_i - O_{def})^2}$$

where i for different sources and def denotes the default. For each source from 1 to 5 above, if there are 2 variations in one source in opposite directions, we keep the variation that result in larger absolute change in <Npart> or <Ncoll> from that of default.

Total Uncertainty

$$\Delta_{total} = \sqrt{(\Delta_{stat})^2 + (\Delta_{sys})^2}$$

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	24.2980	0.0067	0.4037	0.4037	24.3719	23.9275	24.2082	24.3784	24.6761	24.3769	24.2474
5.0	10.0	21.5627	0.0070	0.6843	0.6843	21.7514	20.9490	21.4169	21.6130	22.1745	21.7939	21.5446
10.0	15.0	19.5521	0.0075	0.7642	0.7642	19.7653	18.8917	19.3736	19.5882	20.2350	19.8184	19.5404
15.0	20.0	17.7039	0.0072	0.7767	0.7767	17.8987	17.0188	17.4763	17.7106	18.3856	18.0133	17.6930
20.0	25.0	15.9375	0.0075	0.8694	0.8695	16.1563	15.2990	15.5899	15.9348	16.6402	16.4002	15.9504
25.0	30.0	14.2961	0.0068	0.8258	0.8259	14.4836	13.6982	13.9498	14.2809	14.9445	14.7716	14.3017
30.0	35.0	12.8422	0.0073	0.7962	0.7962	13.0095	12.2894	12.4561	12.8365	13.4370	13.3442	12.8559
35.0	40.0	11.4629	0.0061	0.8401	0.8401	11.6139	10.9713	11.0930	11.4395	12.0040	12.0872	11.4741
40.0	45.0	10.2061	0.0067	0.8106	0.8107	10.3720	9.7663	9.8212	10.1795	10.6985	10.8278	10.2150
45.0	50.0	9.1905	0.0060	0.6805	0.6805	9.3333	8.8009	8.6804	9.1737	9.6173	9.6996	9.1867
50.0	55.0	8.1768	0.0054	0.6572	0.6572	8.3385	7.8345	7.6684	8.1729	8.5556	8.6888	8.1928
55.0	60.0	7.1843	0.0047	0.6113	0.6113	7.3449	6.8909	6.8195	7.1801	7.5068	7.6781	7.1852
60.0	65.0	6.3366	0.0047	0.5759	0.5759	6.5088	6.0959	5.9926	6.3412	6.6084	6.8142	6.3399
65.0	70.0	5.7661	0.0052	0.5216	0.5216	5.9214	5.5620	5.3224	5.7673	5.9921	6.1048	5.7598
70.0	75.0	5.2144	0.0037	0.4888	0.4888	5.3585	5.0472	4.7899	5.2113	5.4091	5.5442	5.2073
75.0	80.0	4.5872	0.0032	0.4539	0.4539	4.7122	4.4452	4.1862	4.5901	4.7363	4.9972	4.5820

Percentage difference from default

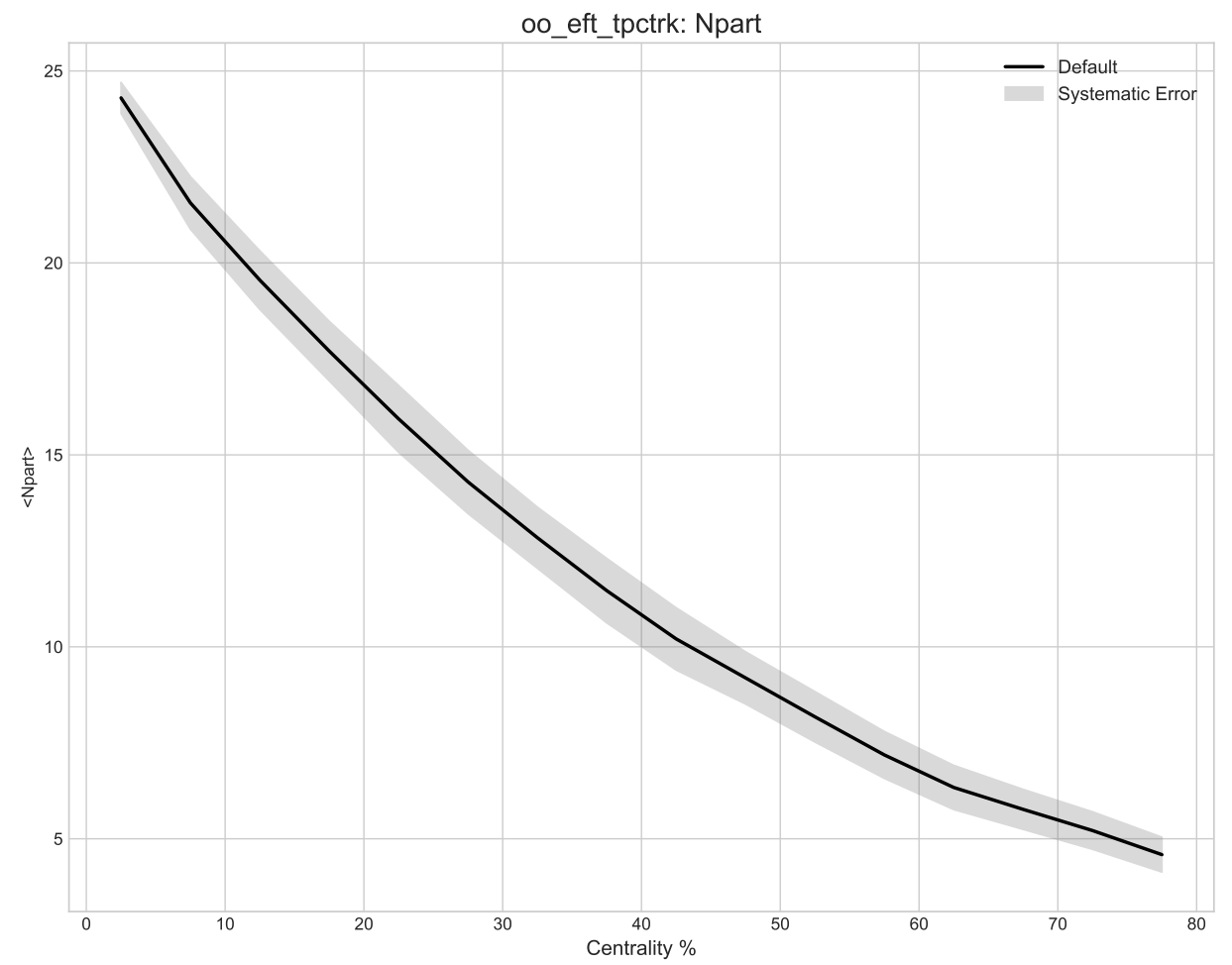
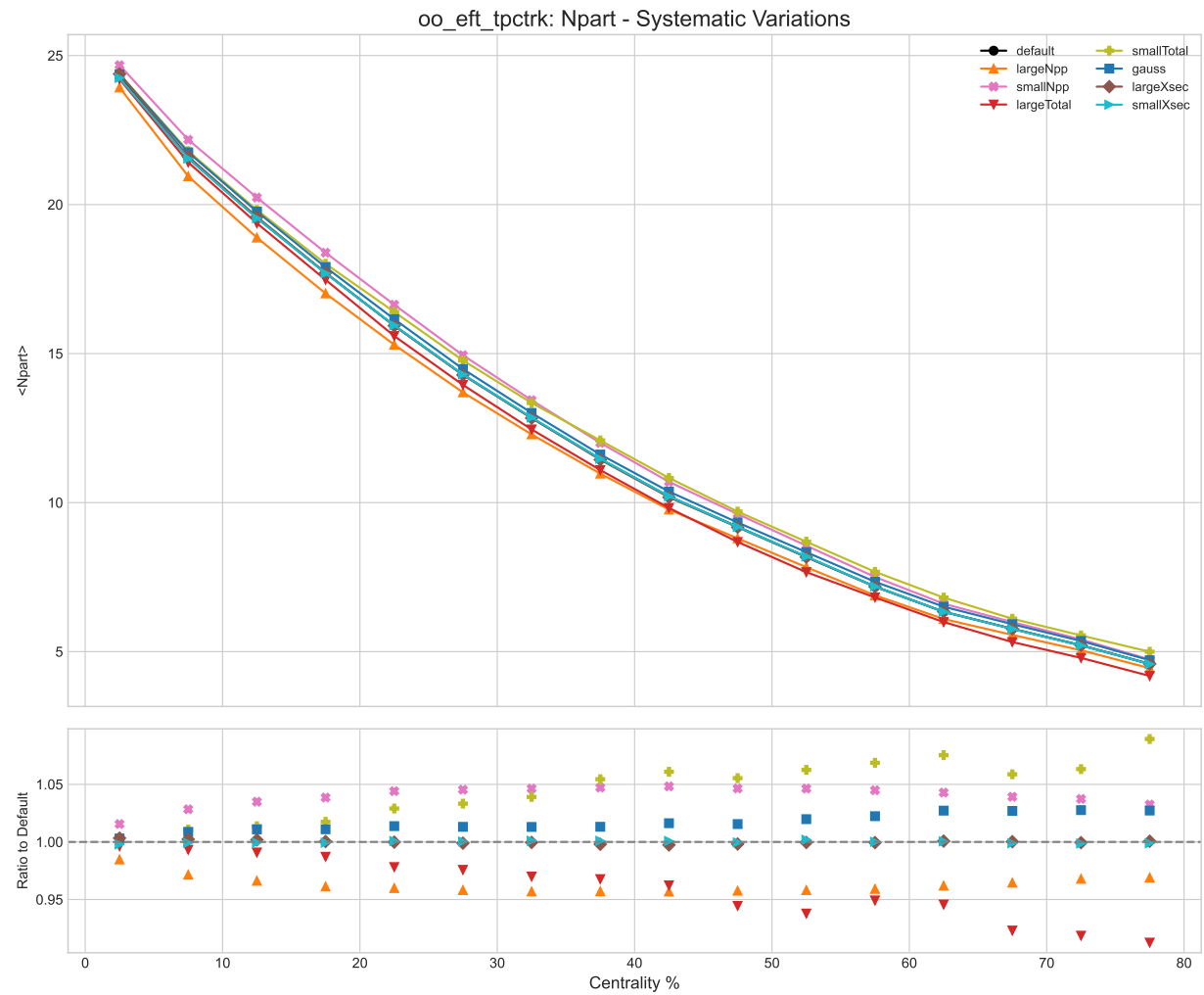
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	1.66	0.30	-1.52	-0.37	0.33	1.56	0.32	-0.21
5.0	10.0	3.17	0.87	-2.85	-0.68	0.23	2.84	1.07	-0.08
10.0	15.0	3.91	1.09	-3.38	-0.91	0.18	3.49	1.36	-0.06
15.0	20.0	4.39	1.10	-3.87	-1.29	0.04	3.85	1.75	-0.06
20.0	25.0	5.46	1.37	-4.01	-2.18	-0.02	4.41	2.90	0.08
25.0	30.0	5.78	1.31	-4.18	-2.42	-0.11	4.54	3.33	0.04
30.0	35.0	6.20	1.30	-4.30	-3.01	-0.04	4.63	3.91	0.11
35.0	40.0	7.33	1.32	-4.29	-3.23	-0.20	4.72	5.45	0.10
40.0	45.0	7.94	1.63	-4.31	-3.77	-0.26	4.82	6.09	0.09
45.0	50.0	7.40	1.55	-4.24	-5.55	-0.18	4.64	5.54	-0.04
50.0	55.0	8.04	1.98	-4.19	-6.22	-0.05	4.63	6.26	0.20
55.0	60.0	8.51	2.24	-4.08	-5.08	-0.06	4.49	6.87	0.01
60.0	65.0	9.09	2.72	-3.80	-5.43	0.07	4.29	7.54	0.05
65.0	70.0	9.05	2.69	-3.54	-7.69	0.02	3.92	5.87	-0.11
70.0	75.0	9.37	2.76	-3.21	-8.14	-0.06	3.73	6.33	-0.14
75.0	80.0	9.89	2.73	-3.10	-8.74	0.06	3.25	8.94	-0.11

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	30.1652	0.0155	1.7200	1.7201	28.7399	29.3875	29.9798	30.6459	30.9785	30.3309	29.7837
5.0	10.0	24.7916	0.0132	1.5795	1.5795	23.7912	23.7772	24.5622	25.1378	25.8898	25.2015	24.5758
10.0	15.0	21.5309	0.0128	1.4585	1.4586	20.6684	20.5134	21.2358	21.7742	22.6057	21.9418	21.3108
15.0	20.0	18.7660	0.0114	1.3617	1.3617	17.9776	17.7718	18.4380	18.9083	19.7657	19.2135	18.5845
20.0	25.0	16.2872	0.0111	1.3294	1.3294	15.6358	15.4256	15.8183	16.4088	17.2567	16.9102	16.1812
25.0	30.0	14.1144	0.0096	1.2148	1.2149	13.5222	13.3461	13.6637	14.2001	14.9677	14.7371	14.0182
30.0	35.0	12.2725	0.0098	1.1106	1.1106	11.7559	11.5962	11.8108	12.3388	13.0256	12.9009	12.2080
35.0	40.0	10.6149	0.0078	1.0872	1.0873	10.1547	10.0347	10.1762	10.6417	11.2548	11.3613	10.5542
40.0	45.0	9.1575	0.0081	0.9791	0.9792	8.7963	8.6542	8.7215	9.1752	9.7234	9.8687	9.1103
45.0	50.0	8.0204	0.0071	0.8086	0.8087	7.7036	7.5901	7.4657	8.0390	8.4979	8.5887	7.9713
50.0	55.0	6.9298	0.0061	0.7188	0.7188	6.6963	6.5683	6.3985	6.9527	7.3393	7.4720	6.9087
55.0	60.0	5.8949	0.0051	0.6312	0.6312	5.7282	5.5965	5.5256	5.9089	6.2267	6.4049	5.8722
60.0	65.0	5.0484	0.0049	0.5526	0.5527	4.9422	4.8060	4.7068	5.0649	5.3153	5.5201	5.0291
65.0	70.0	4.4886	0.0052	0.4833	0.4834	4.4078	4.2903	4.0680	4.5044	4.7118	4.8184	4.4691
70.0	75.0	3.9631	0.0037	0.4404	0.4404	3.9094	3.8054	3.5676	3.9703	4.1480	4.2744	3.9429
75.0	80.0	3.3814	0.0030	0.4046	0.4046	3.3540	3.2517	3.0170	3.3918	3.5196	3.7604	3.3674

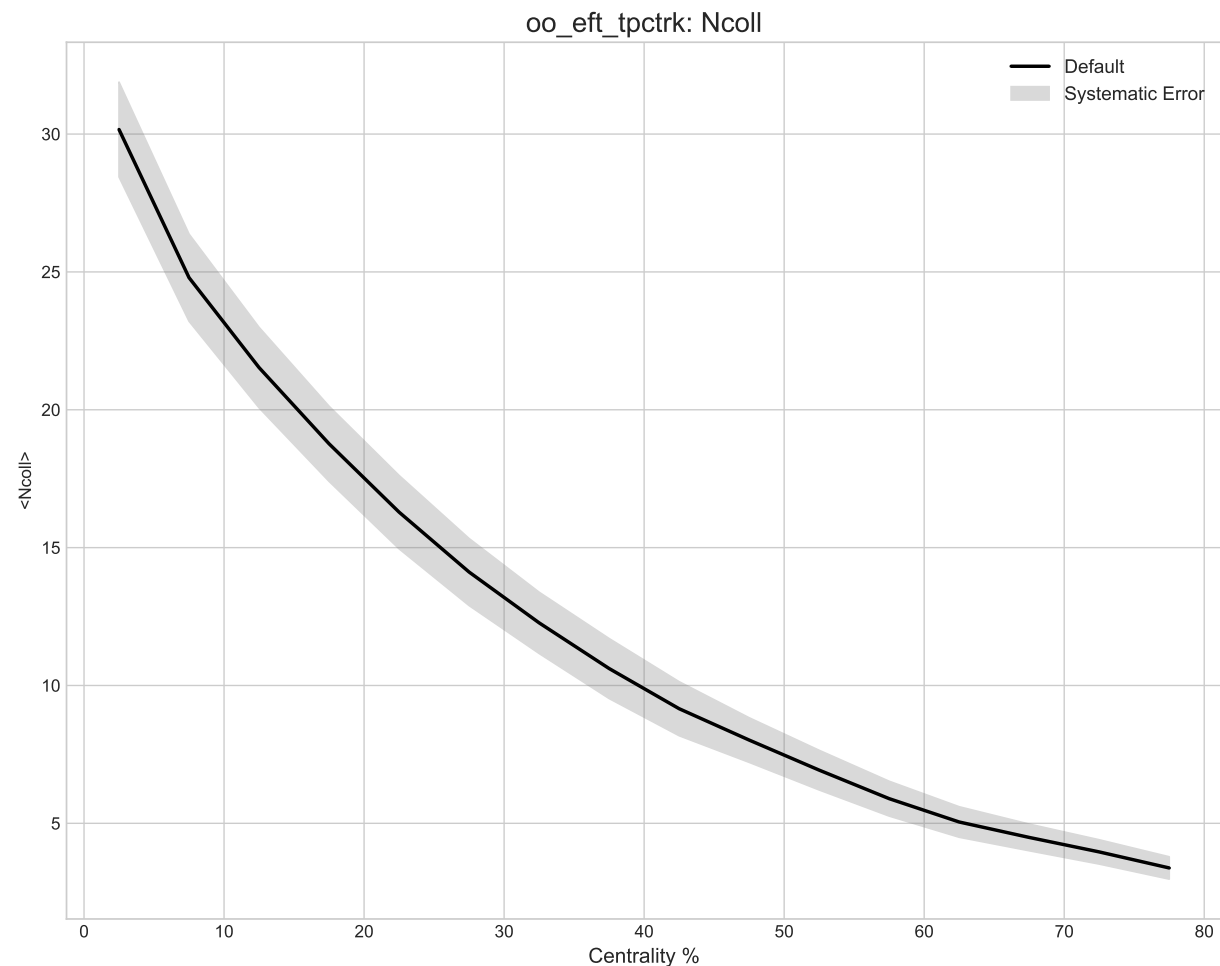
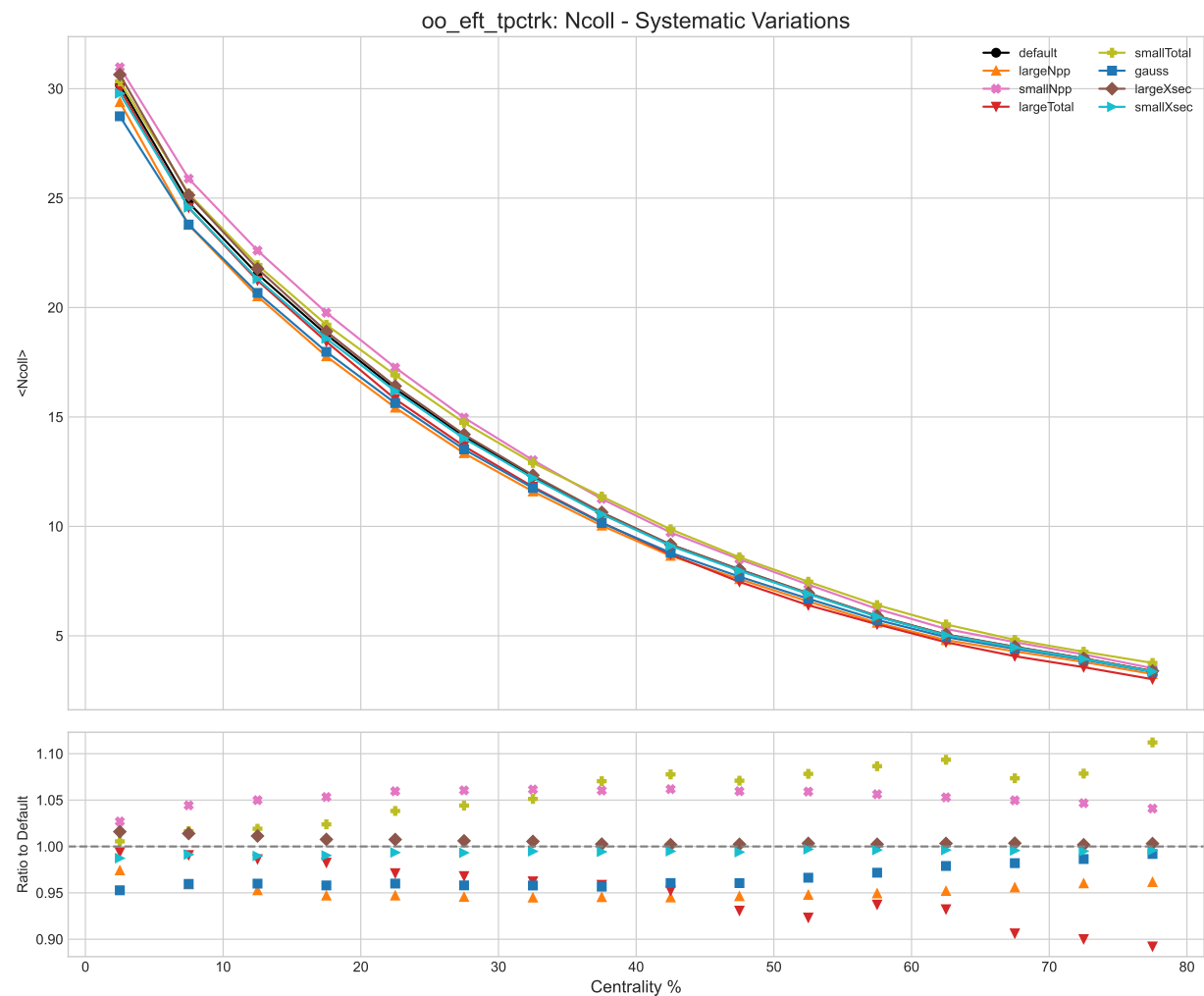
Percentage difference from default

min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	5.70	-4.72	-2.58	-0.61	1.59	2.70	0.55	-1.26
5.0	10.0	6.37	-4.04	-4.09	-0.93	1.40	4.43	1.65	-0.87
10.0	15.0	6.77	-4.01	-4.73	-1.37	1.13	4.99	1.91	-1.02
15.0	20.0	7.26	-4.20	-5.30	-1.75	0.76	5.33	2.38	-0.97
20.0	25.0	8.16	-4.00	-5.29	-2.88	0.75	5.95	3.82	-0.65
25.0	30.0	8.61	-4.20	-5.44	-3.19	0.61	6.05	4.41	-0.68
30.0	35.0	9.05	-4.21	-5.51	-3.76	0.54	6.14	5.12	-0.53
35.0	40.0	10.24	-4.33	-5.47	-4.13	0.25	6.03	7.03	-0.57
40.0	45.0	10.69	-3.94	-5.50	-4.76	0.19	6.18	7.77	-0.52
45.0	50.0	10.08	-3.95	-5.36	-6.92	0.23	5.95	7.09	-0.61
50.0	55.0	10.37	-3.37	-5.22	-7.67	0.33	5.91	7.82	-0.30
55.0	60.0	10.71	-2.83	-5.06	-6.27	0.24	5.63	8.65	-0.39
60.0	65.0	10.95	-2.10	-4.80	-6.77	0.33	5.29	9.34	-0.38
65.0	70.0	10.77	-1.80	-4.42	-9.37	0.35	4.97	7.35	-0.43
70.0	75.0	11.11	-1.36	-3.98	-9.98	0.18	4.66	7.85	-0.51
75.0	80.0	11.97	-0.81	-3.84	-10.78	0.31	4.09	11.21	-0.41

<Npart> vs centrality



<Ncoll> vs centrality



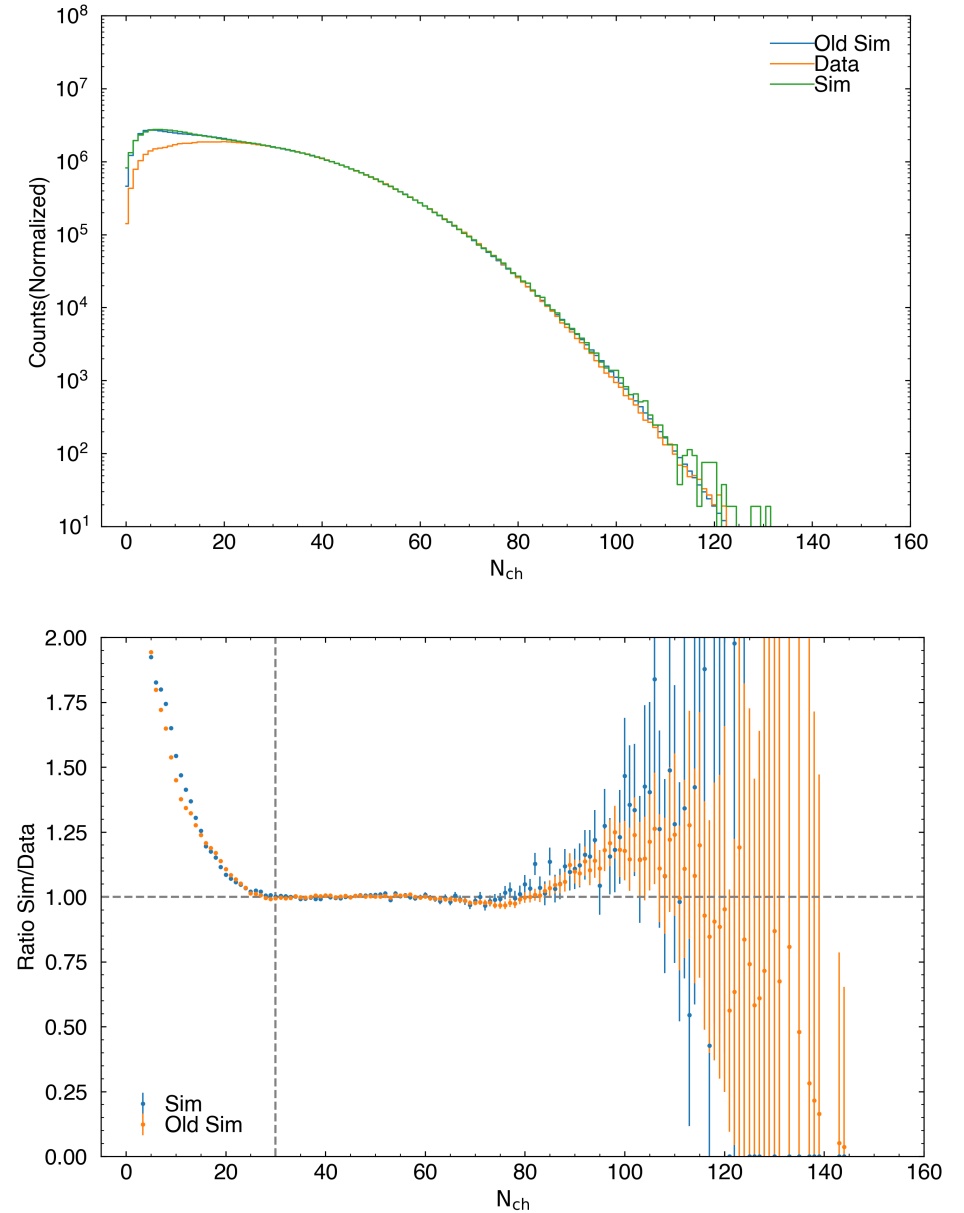
d+Au

d+Au Nch cuts

Percentile(%)	Old cuts	Sim
0 - 1	69.37	68.36
1 - 2	63.74	62.72
2 - 3	60.18	59.13
3 - 4	57.50	56.42
4 - 5	55.31	54.21
5 - 6	53.45	52.34
6 - 7	51.81	50.71
7 - 8	50.34	49.24
8 - 9	49.00	47.89
9 - 10	47.77	46.66
10 - 15	42.69	41.53
15 - 20	38.66	37.45
20 - 25	35.19	33.96
25 - 30	32.08	30.85
30 - 35	29.22	27.99
35 - 40	26.55	25.34
40 - 45	24.06	22.83
45 - 50	21.74	20.45
50 - 55	19.54	18.21
55 - 60	17.45	16.07
60 - 65	15.43	14.07
65 - 70	13.46	12.17
70 - 75	11.55	10.35
75 - 80	9.70	8.61
80 - 85	7.93	6.90
85 - 90	6.22	5.17
90 - 95	4.42	3.27
95 - 100	0.00	0.00

Glauber Fit Parameters:

Sim: npp=4.7, k=6.11, x=0.14, eff=0.14



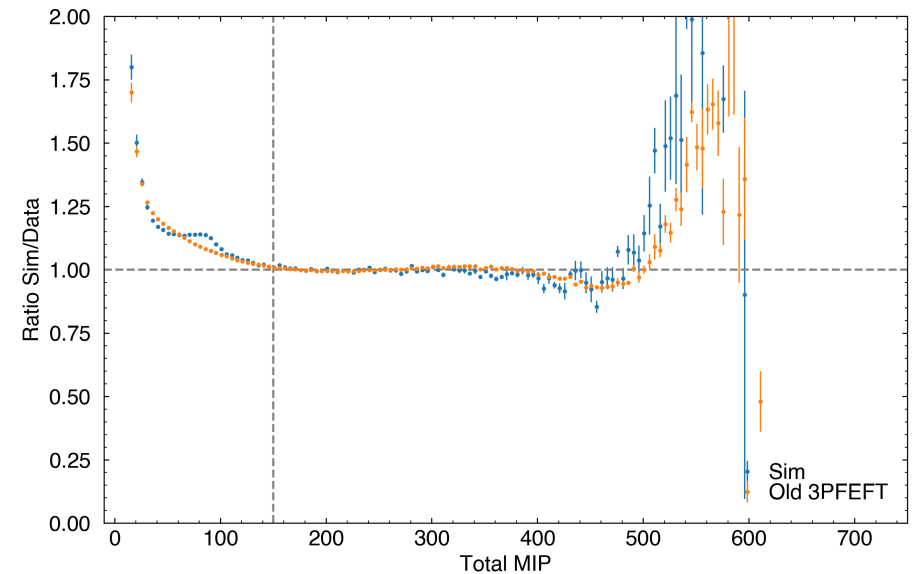
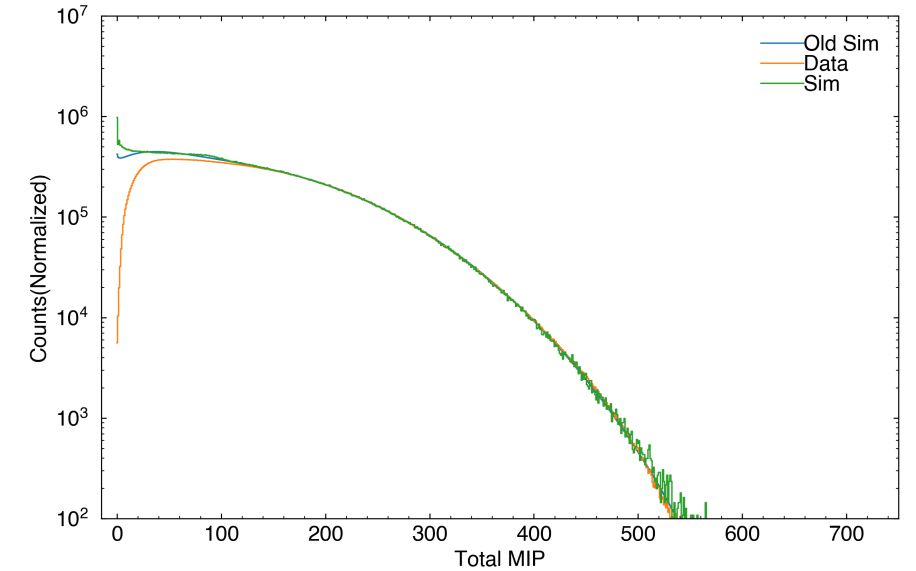
Fit range > 30

d+Au Total MIP cuts

Percentile(%)	Old cuts	Sim
0 - 1	364.37	362.77
1 - 2	334.21	332.02
2 - 3	314.85	312.39
3 - 4	300.20	297.54
4 - 5	288.23	285.01
5 - 6	278.01	274.79
6 - 7	269.04	265.76
7 - 8	260.99	257.71
8 - 9	253.67	250.41
9 - 10	246.92	243.65
10 - 15	219.00	215.67
15 - 20	196.90	193.48
20 - 25	178.10	174.55
25 - 30	161.44	157.79
30 - 35	146.28	142.49
35 - 40	132.24	128.30
40 - 45	119.08	115.06
45 - 50	106.64	102.59
50 - 55	94.81	91.03
55 - 60	83.52	80.07
60 - 65	72.70	69.32
65 - 70	62.28	58.68
70 - 75	52.17	48.19
75 - 80	42.28	37.82
80 - 85	32.46	27.59
85 - 90	22.51	17.55
90 - 95	12.08	7.82
95 - 100	0.00	0.00

Glauber Fit Parameters:

Sim: npp=32.67, k=0.74, x=0.14, eff=0.14



Fit range > 150

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	20.1141	0.0074	0.5467	0.5467	19.9959	19.9536	19.7082	20.1100	20.2985	20.2597	20.5468	20.2814	19.9045
5.0	10.0	17.4876	0.0068	0.5605	0.5605	17.4789	17.4058	16.9871	17.3523	17.6176	17.5541	18.0093	17.5934	17.3591
10.0	15.0	16.0804	0.0066	0.6043	0.6044	16.1034	16.0132	15.5841	15.8371	16.1708	16.1362	16.6215	16.0895	15.9921
15.0	20.0	14.7651	0.0057	0.5705	0.5705	14.8145	14.7189	14.2710	14.6504	14.8472	14.8033	15.3026	14.9236	14.7031
20.0	25.0	13.6722	0.0066	0.5888	0.5888	13.7266	13.6250	13.1648	13.5028	13.7106	13.6808	14.1832	13.9486	13.6097
25.0	30.0	12.7745	0.0062	0.6004	0.6004	12.8406	12.7509	12.2959	12.4630	12.8176	12.7962	13.2811	13.0775	12.7307
30.0	35.0	11.8366	0.0058	0.5834	0.5834	11.9144	11.8315	11.3797	11.5149	11.8686	11.8323	12.3159	12.1480	11.8101
35.0	40.0	10.8568	0.0054	0.5769	0.5769	10.9351	10.8648	10.4267	10.5360	10.8816	10.8494	11.3227	11.1869	10.8432
40.0	45.0	10.0224	0.0063	0.5620	0.5621	10.0849	10.0340	9.6211	9.6752	10.0341	10.0056	10.4592	10.3673	10.0035
45.0	50.0	9.1393	0.0048	0.6830	0.6830	9.1960	9.1678	8.7589	8.7887	9.1505	9.1202	9.5550	9.6773	9.1416
50.0	55.0	8.2408	0.0055	0.6753	0.6753	8.3108	8.2762	7.8954	7.8814	8.2469	8.2005	8.6308	8.7861	8.2329
55.0	60.0	7.5127	0.0051	0.5154	0.5154	7.5725	7.5457	7.1916	7.1375	7.5156	7.4754	7.8589	7.8759	7.5079
60.0	65.0	6.7718	0.0047	0.4962	0.4962	6.8541	6.8181	6.4915	6.4167	6.7791	6.7478	7.0858	7.1443	6.7702
65.0	70.0	6.0667	0.0042	0.4524	0.4524	6.1459	6.1017	5.8139	5.7116	6.0614	6.0173	6.3308	6.4194	6.0582
70.0	75.0	5.3725	0.0037	0.5704	0.5704	5.4646	5.4165	5.1766	5.0464	5.3758	5.3323	5.5979	5.8864	5.3789
75.0	80.0	4.8878	0.0048	0.5370	0.5370	4.9897	4.9287	4.7171	4.4181	4.8911	4.8560	5.0771	5.3782	4.8831

Percentage difference from default

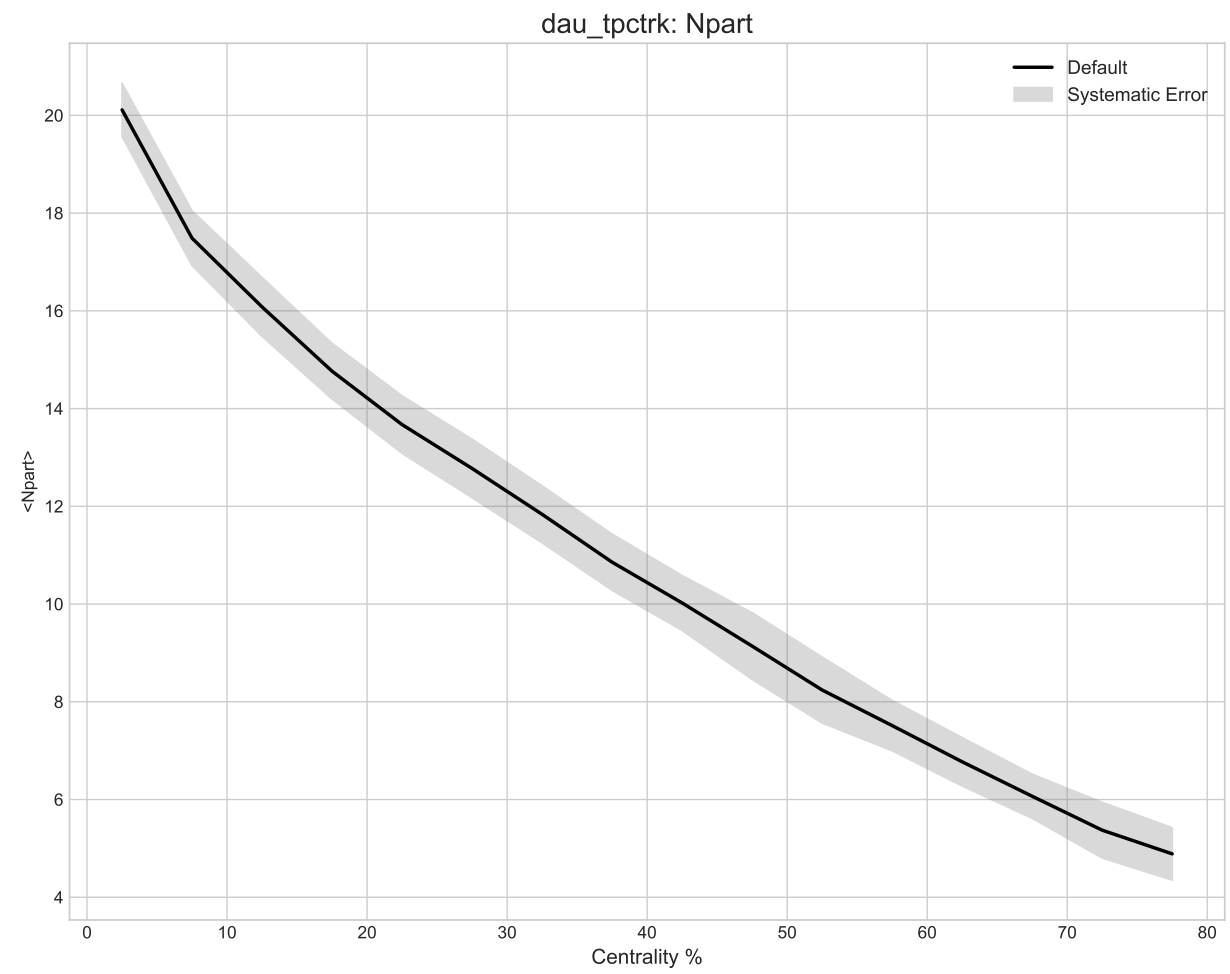
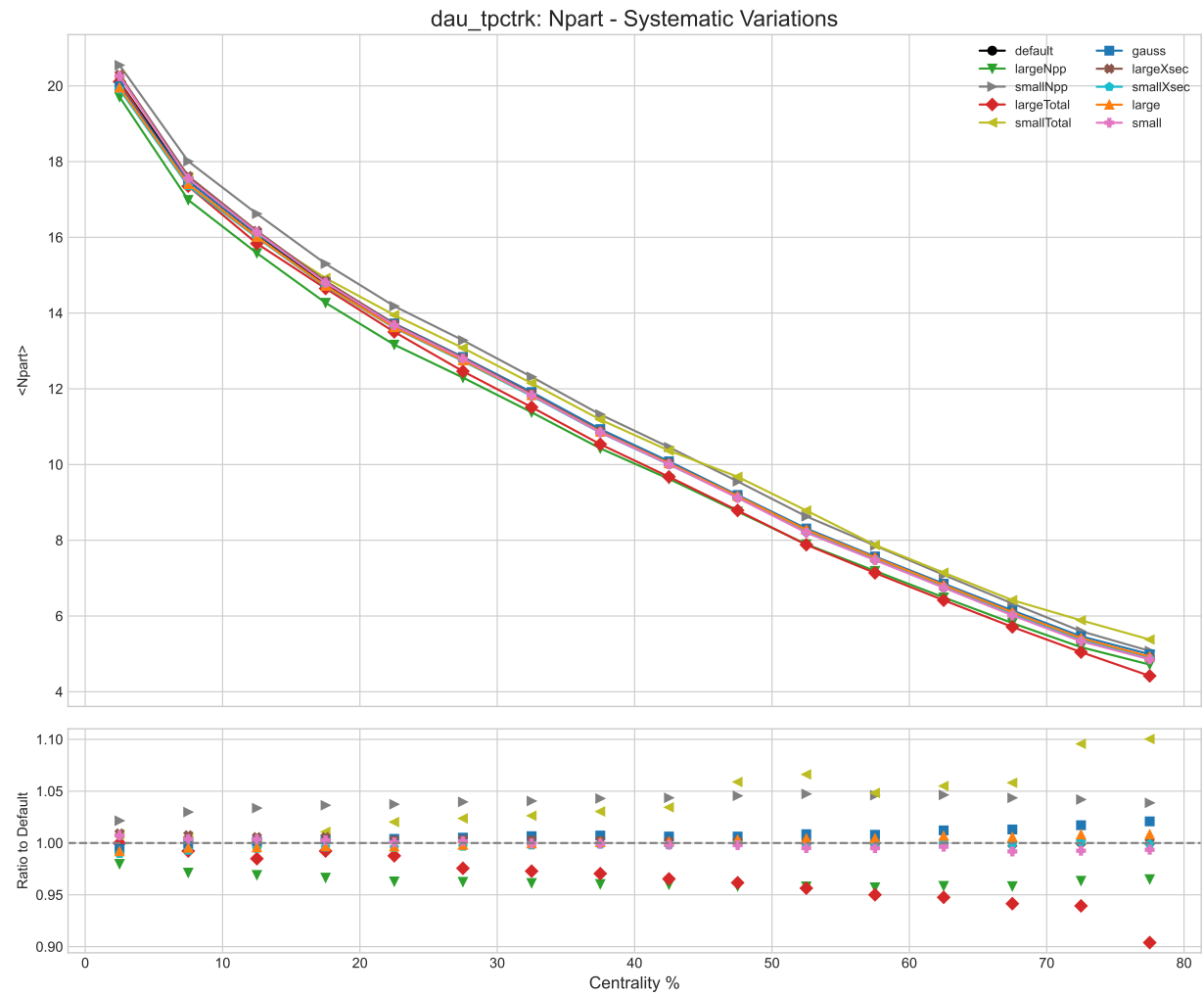
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	2.72	-0.59	-0.80	-2.02	-0.02	0.92	0.72	2.15	0.83	-1.04
5.0	10.0	3.20	-0.05	-0.47	-2.86	-0.77	0.74	0.38	2.98	0.60	-0.74
10.0	15.0	3.76	0.14	-0.42	-3.09	-1.51	0.56	0.35	3.37	0.06	-0.55
15.0	20.0	3.86	0.33	-0.31	-3.35	-0.78	0.56	0.26	3.64	1.07	-0.42
20.0	25.0	4.31	0.40	-0.34	-3.71	-1.24	0.28	0.06	3.74	2.02	-0.46
25.0	30.0	4.70	0.52	-0.19	-3.75	-2.44	0.34	0.17	3.97	2.37	-0.34
30.0	35.0	4.93	0.66	-0.04	-3.86	-2.72	0.27	-0.04	4.05	2.63	-0.22
35.0	40.0	5.31	0.72	0.07	-3.96	-2.95	0.23	-0.07	4.29	3.04	-0.13
40.0	45.0	5.61	0.62	0.12	-4.00	-3.46	0.12	-0.17	4.36	3.44	-0.19
45.0	50.0	7.47	0.62	0.31	-4.16	-3.84	0.12	-0.21	4.55	5.89	0.03
50.0	55.0	8.19	0.85	0.43	-4.19	-4.36	0.07	-0.49	4.73	6.62	-0.10
55.0	60.0	6.86	0.80	0.44	-4.28	-5.00	0.04	-0.50	4.61	4.83	-0.06
60.0	65.0	7.33	1.22	0.68	-4.14	-5.24	0.11	-0.36	4.64	5.50	-0.02
65.0	70.0	7.46	1.31	0.58	-4.17	-5.85	-0.09	-0.82	4.35	5.81	-0.14
70.0	75.0	10.62	1.72	0.82	-3.64	-6.07	0.06	-0.75	4.20	9.57	0.12
75.0	80.0	10.99	2.09	0.84	-3.49	-9.61	0.07	-0.65	3.87	10.03	-0.10

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	19.6465	0.0082	0.5593	0.5593	19.5901	19.4557	19.2507	19.6385	19.8752	19.8256	20.0706	19.8123	19.3899
5.0	10.0	17.0484	0.0078	0.5995	0.5995	16.9120	16.9473	16.5539	16.9166	17.2352	17.1472	17.5744	17.1591	16.8573
10.0	15.0	15.6303	0.0076	0.6565	0.6566	15.4324	15.5509	15.1215	15.3828	15.7798	15.7085	16.1798	15.6354	15.4854
15.0	20.0	14.2845	0.0066	0.6333	0.6334	14.0416	14.2244	13.7597	14.1525	14.4156	14.3271	14.8306	14.4365	14.1657
20.0	25.0	13.1265	0.0077	0.6804	0.6805	12.8659	13.0892	12.5980	12.9535	13.2225	13.1421	13.6683	13.4264	13.0252
25.0	30.0	12.1765	0.0072	0.6836	0.6837	11.9163	12.1616	11.6720	11.8531	12.2683	12.2044	12.7078	12.5054	12.0957
30.0	35.0	11.1760	0.0068	0.6776	0.6776	10.9116	11.1791	10.6809	10.8338	11.2486	11.1748	11.6926	11.5079	11.1185
35.0	40.0	10.1208	0.0063	0.6746	0.6747	9.8705	10.1421	9.6636	9.7809	10.1840	10.1132	10.6275	10.4831	10.0770
40.0	45.0	9.2277	0.0072	0.6582	0.6582	8.9696	9.2485	8.7948	8.8497	9.2695	9.2029	9.6979	9.5969	9.1829
45.0	50.0	8.2825	0.0055	0.7658	0.7659	8.0365	8.3170	7.8662	7.9002	8.3114	8.2487	8.7243	8.8558	8.2602
50.0	55.0	7.3125	0.0062	0.7456	0.7456	7.1161	7.3618	6.9392	6.9303	7.3377	7.2597	7.7294	7.8957	7.2883
55.0	60.0	6.5350	0.0057	0.5739	0.5739	6.3589	6.5851	6.1982	6.1398	6.5506	6.4860	6.9082	6.9241	6.5183
60.0	65.0	5.7541	0.0051	0.5367	0.5368	5.6299	5.8149	5.4624	5.3819	5.7728	5.7174	6.0879	6.1506	5.7362
65.0	70.0	5.0187	0.0046	0.4742	0.4743	4.9248	5.0651	4.7569	4.6503	5.0202	4.9576	5.2948	5.3857	4.9995
70.0	75.0	4.3067	0.0040	0.5778	0.5778	4.2599	4.3609	4.1027	3.9762	4.3132	4.2536	4.5353	4.8325	4.3075
75.0	80.0	3.8154	0.0049	0.5350	0.5350	3.8033	3.8655	3.6427	3.3482	3.8201	3.7753	4.0090	4.3114	3.8077

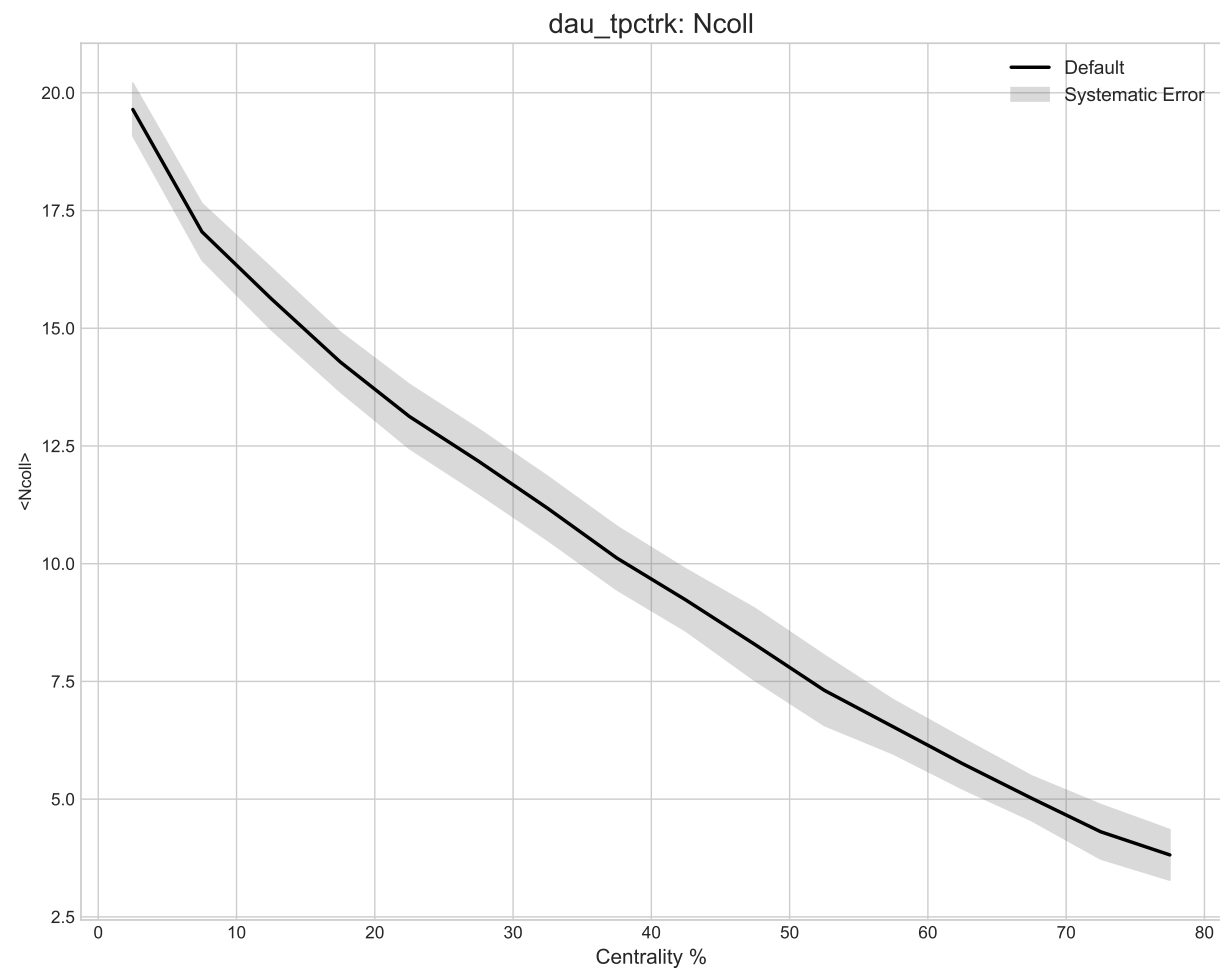
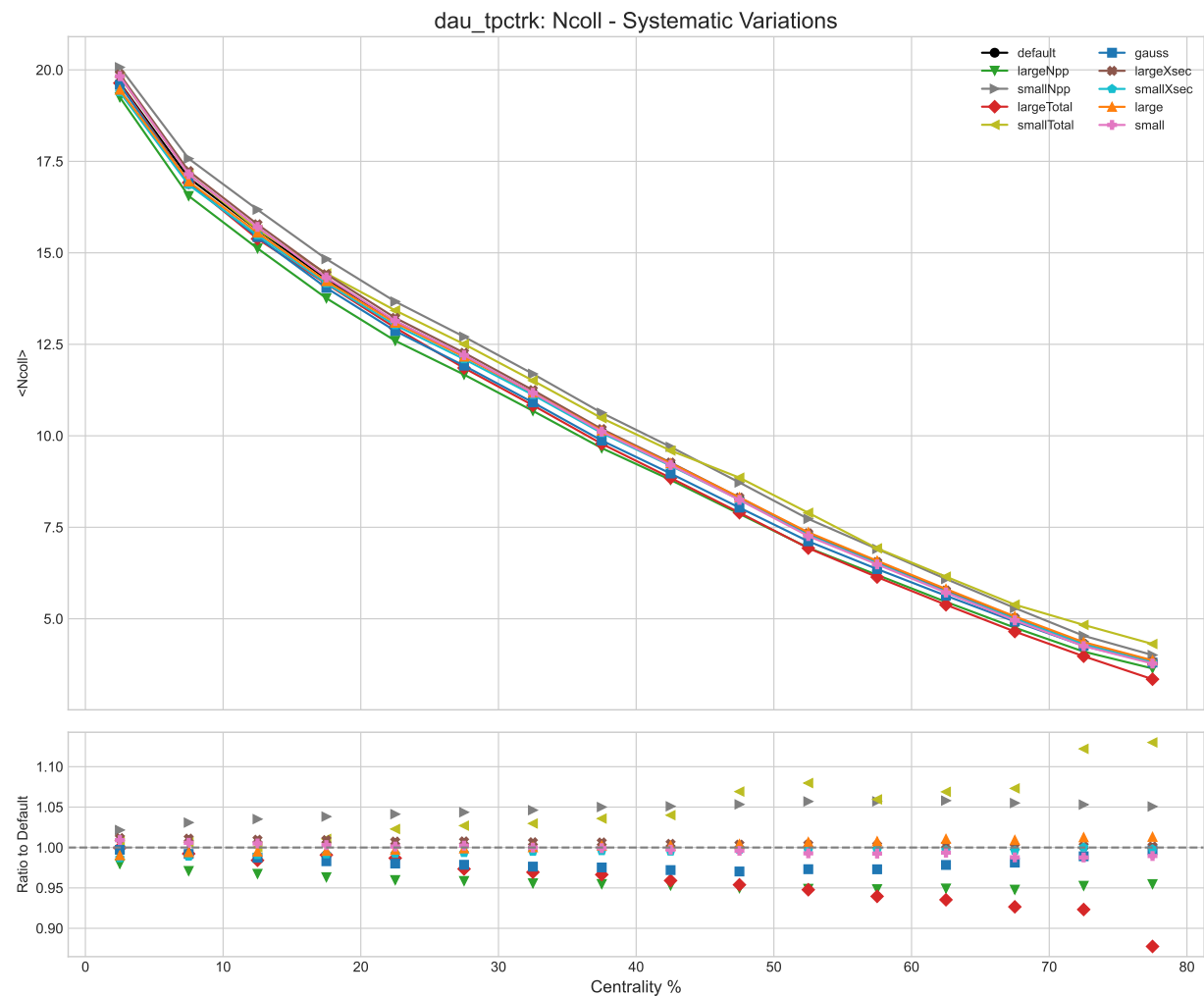
Percentage difference from default

min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	2.85	-0.29	-0.97	-2.01	-0.04	1.16	0.91	2.16	0.84	-1.31
5.0	10.0	3.52	-0.80	-0.59	-2.90	-0.77	1.10	0.58	3.08	0.65	-1.12
10.0	15.0	4.20	-1.27	-0.51	-3.26	-1.58	0.96	0.50	3.52	0.03	-0.93
15.0	20.0	4.43	-1.70	-0.42	-3.67	-0.92	0.92	0.30	3.82	1.06	-0.83
20.0	25.0	5.18	-1.99	-0.28	-4.03	-1.32	0.73	0.12	4.13	2.28	-0.77
25.0	30.0	5.61	-2.14	-0.12	-4.14	-2.66	0.75	0.23	4.36	2.70	-0.66
30.0	35.0	6.06	-2.37	0.03	-4.43	-3.06	0.65	-0.01	4.62	2.97	-0.51
35.0	40.0	6.67	-2.47	0.21	-4.52	-3.36	0.62	-0.07	5.01	3.58	-0.43
40.0	45.0	7.13	-2.80	0.23	-4.69	-4.10	0.45	-0.27	5.10	4.00	-0.49
45.0	50.0	9.25	-2.97	0.42	-5.03	-4.61	0.35	-0.41	5.34	6.92	-0.27
50.0	55.0	10.20	-2.69	0.68	-5.11	-5.23	0.35	-0.72	5.70	7.98	-0.33
55.0	60.0	8.78	-2.70	0.77	-5.15	-6.05	0.24	-0.75	5.71	5.95	-0.26
60.0	65.0	9.33	-2.16	1.06	-5.07	-6.47	0.32	-0.64	5.80	6.89	-0.31
65.0	70.0	9.45	-1.87	0.92	-5.22	-7.34	0.03	-1.22	5.50	7.31	-0.38
70.0	75.0	13.42	-1.09	1.26	-4.74	-7.67	0.15	-1.23	5.31	12.21	0.02
75.0	80.0	14.02	-0.32	1.31	-4.53	-12.24	0.12	-1.05	5.07	13.00	-0.20

<Npart> vs centrality



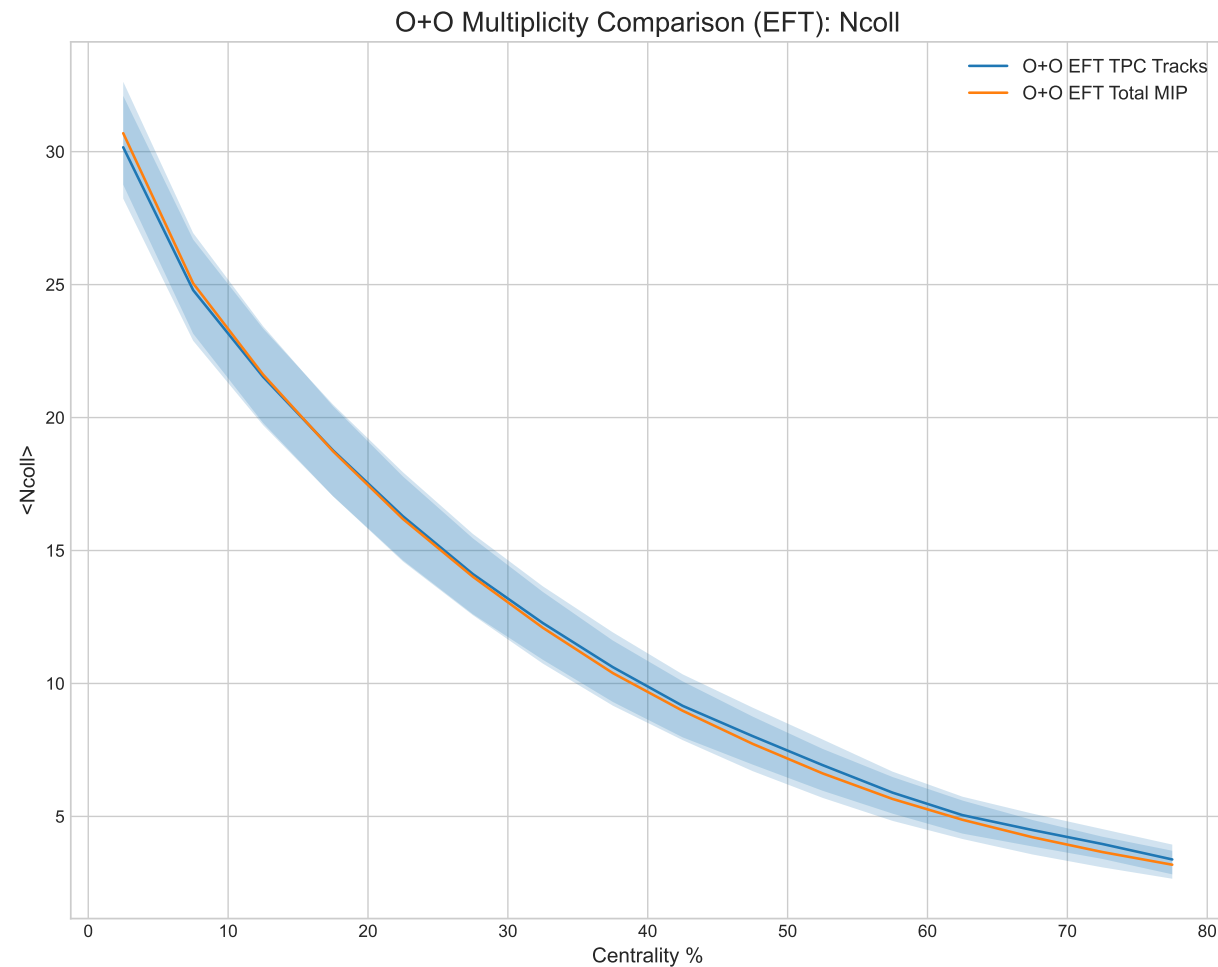
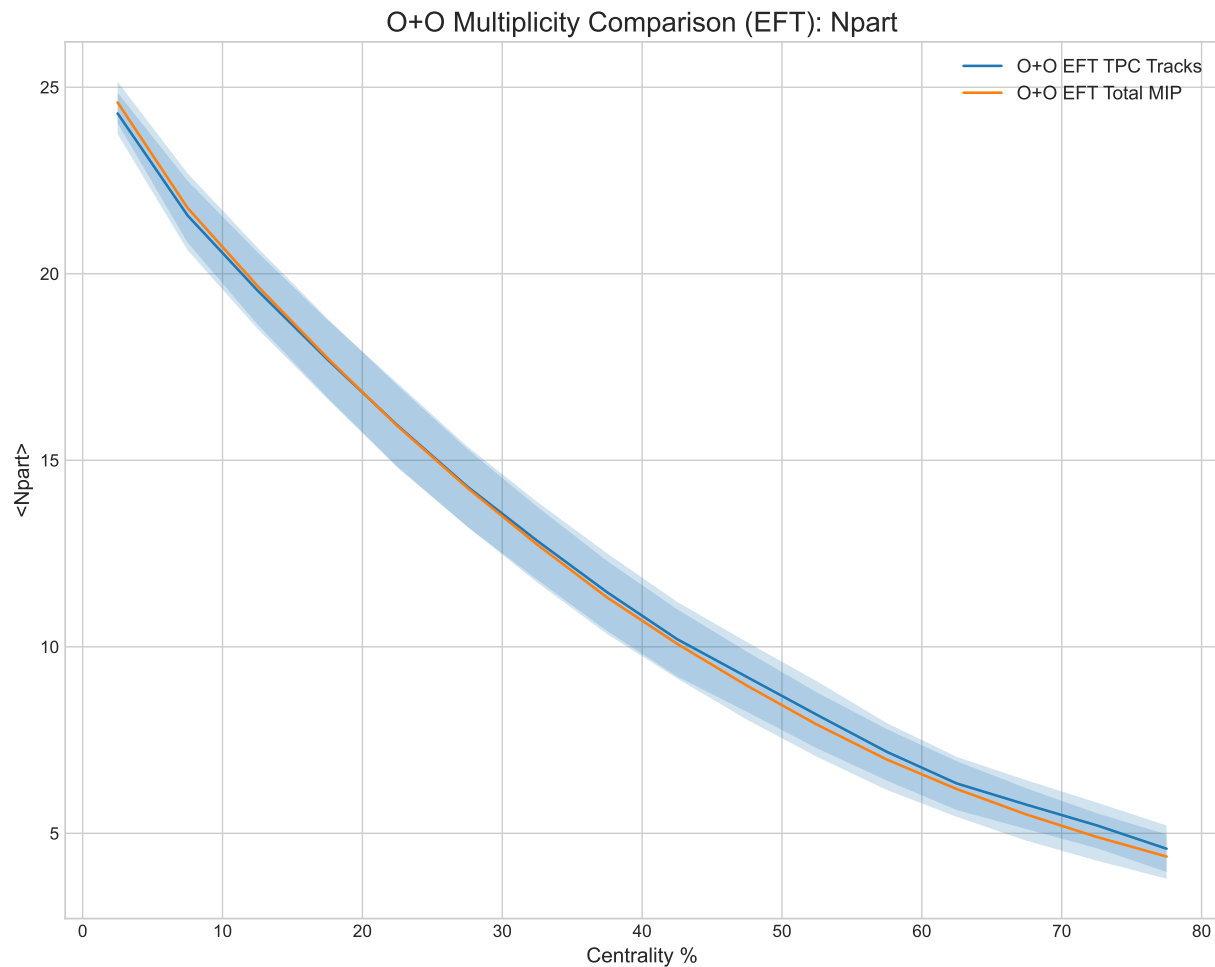
<Ncoll> vs centrality



<Npart> and <Ncoll> vs.
1. Nch or Total MIP
2. Initial Models (for O+O)

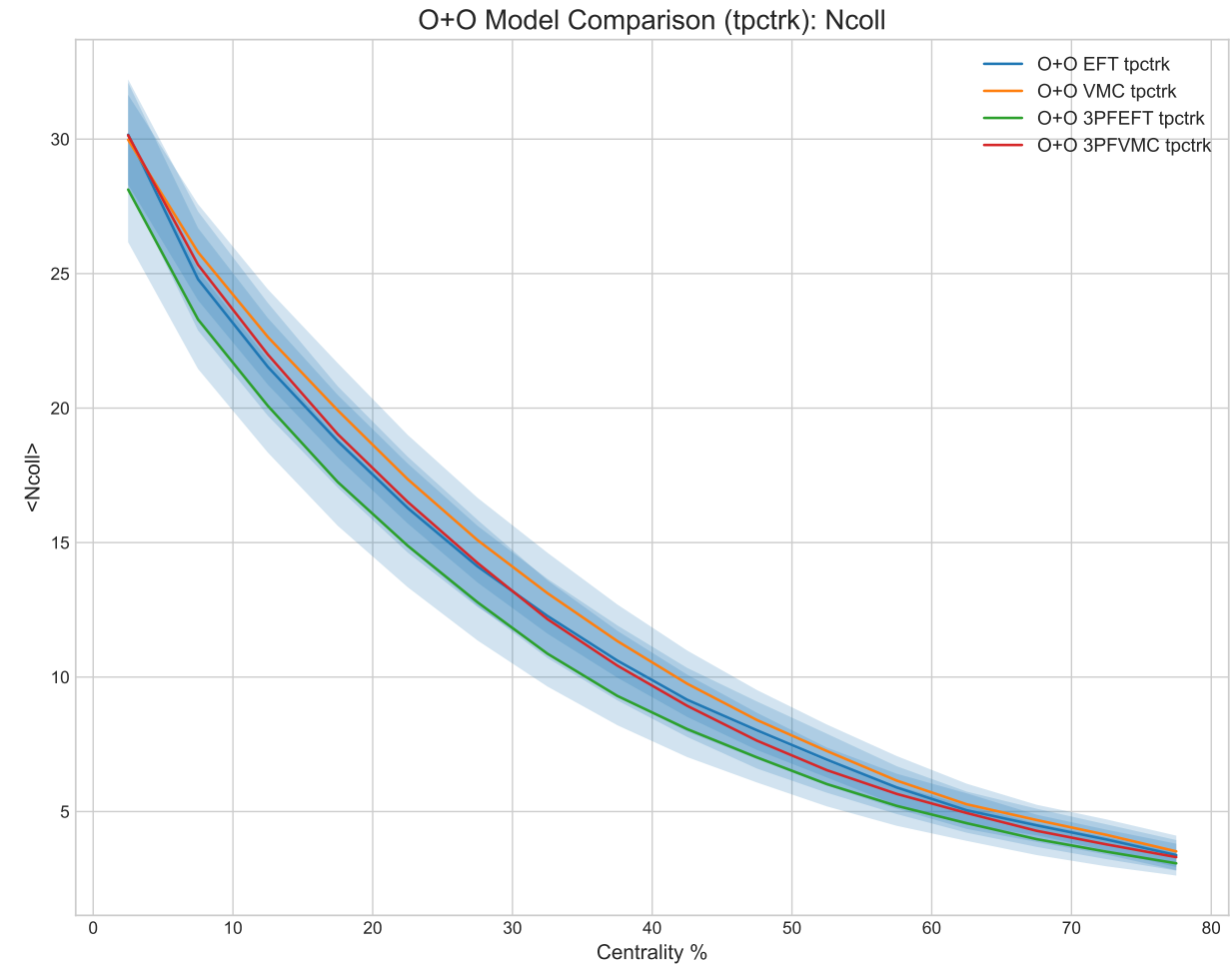
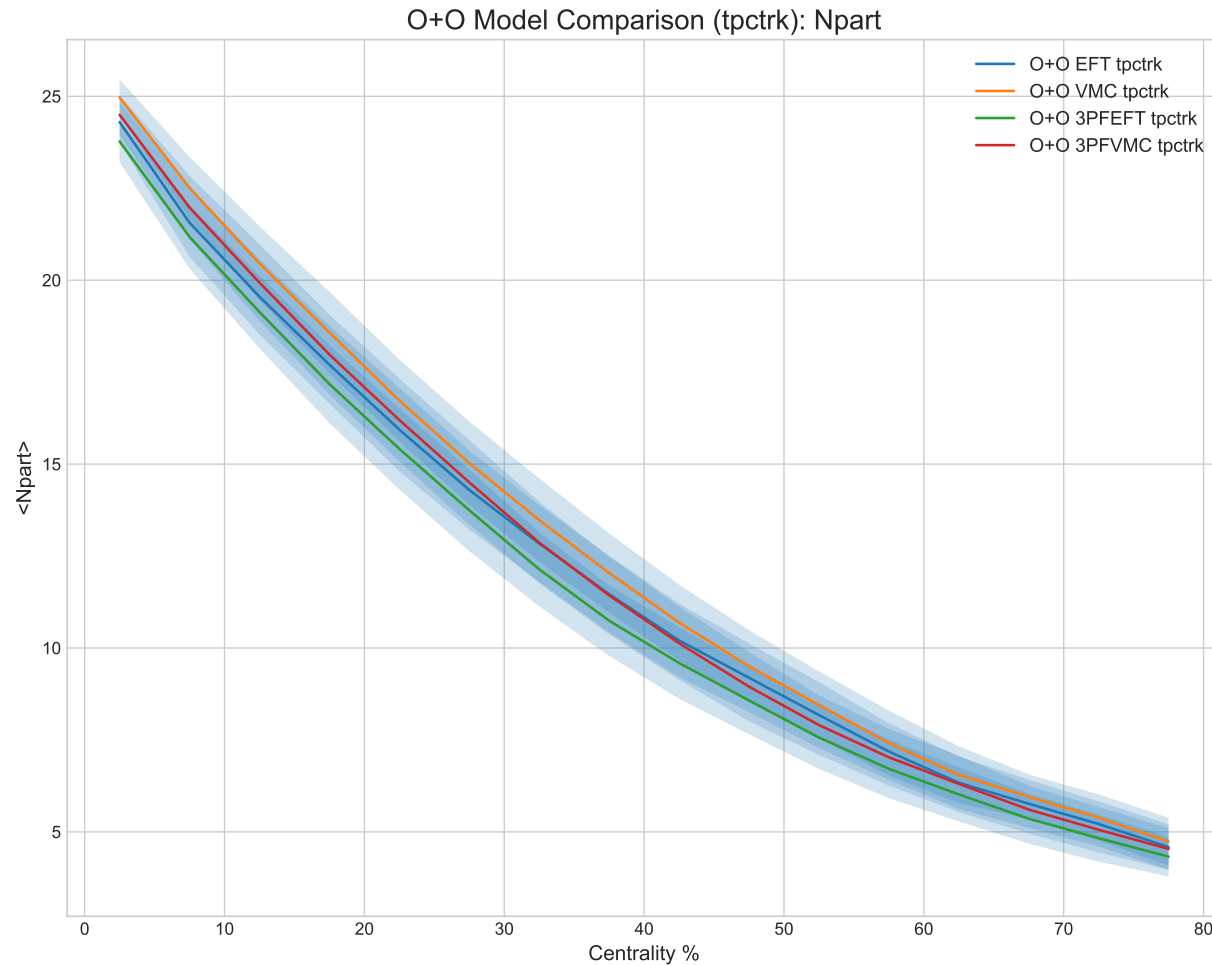
O+O multiplicity used in centrality cut vs. $\langle N_{\text{part}} \rangle$ and $\langle N_{\text{coll}} \rangle$

Due to small difference we see between different multiplicities (Nch vs. Total MIP), there is no preference shown for which one to use for the $\langle N_{\text{part}} \rangle$ and $\langle N_{\text{coll}} \rangle$. We use Nch to align more with the previous centrality definitions in other systems with TPC charged particles.



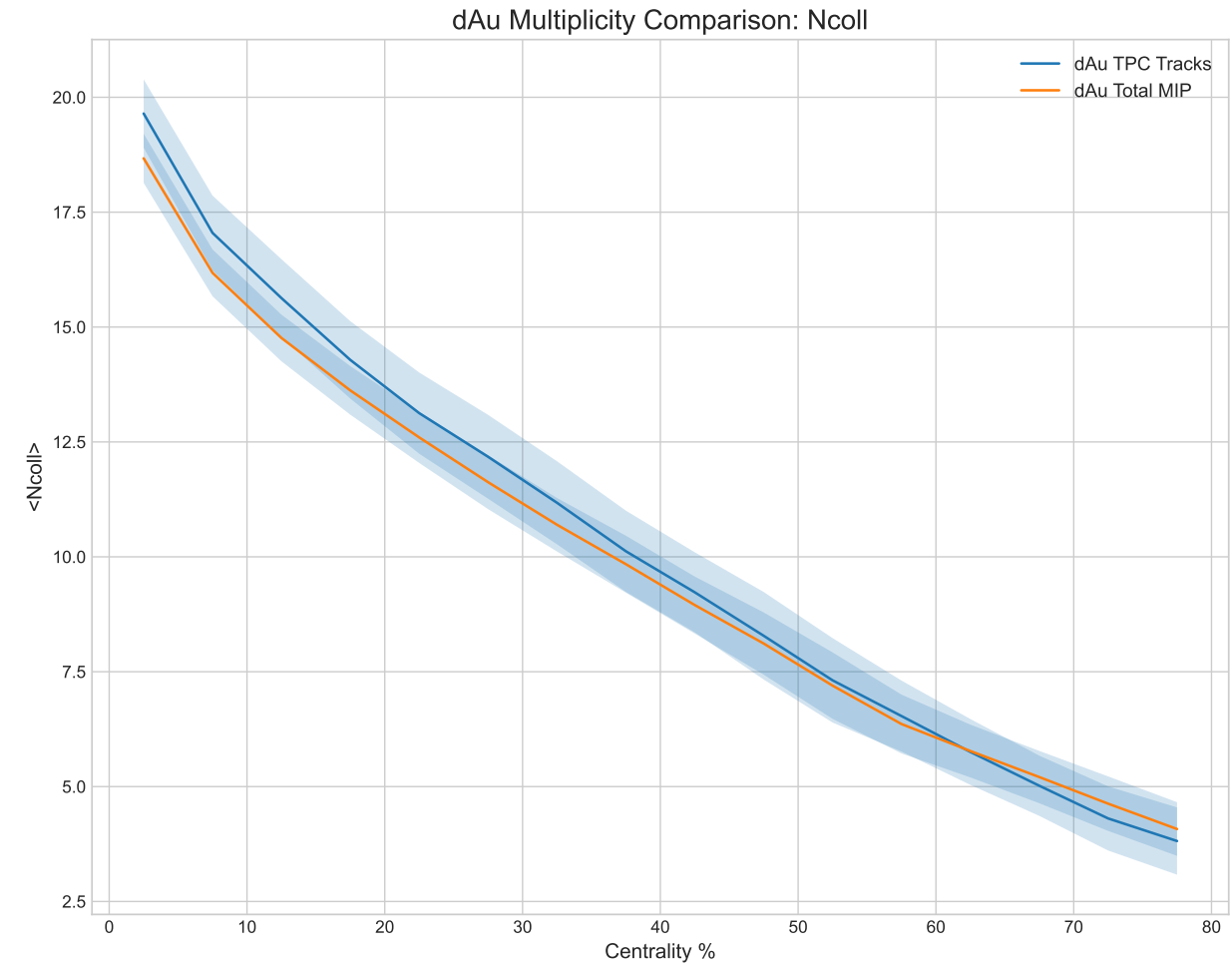
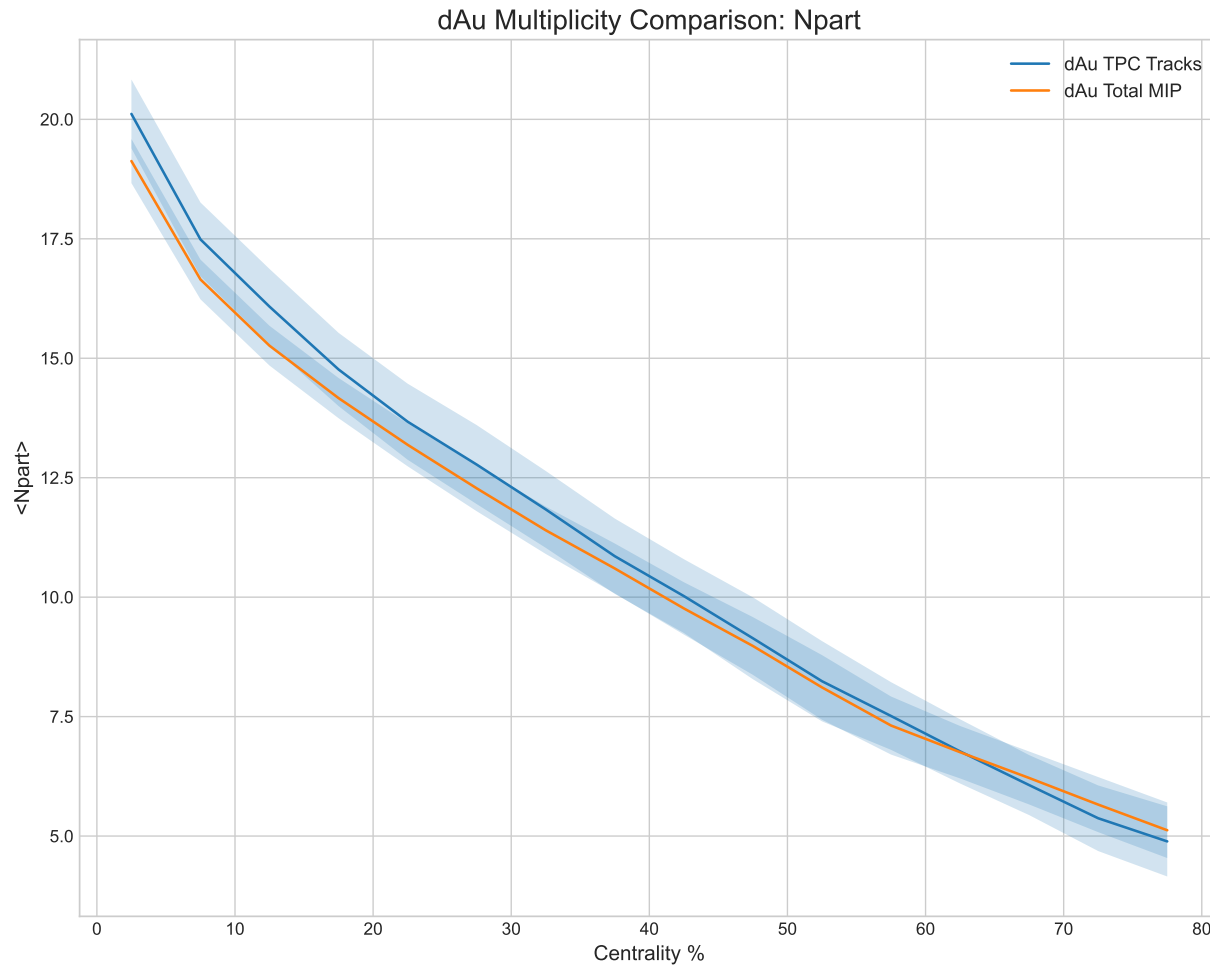
O+O model used in centrality cut vs. $\langle N_{\text{part}} \rangle$ and $\langle N_{\text{coll}} \rangle$

Due to small difference we see between different Models, there is no preference shown for which one to use for the $\langle N_{\text{part}} \rangle$ and $\langle N_{\text{coll}} \rangle$. We use Nch, and the widely used advanced ab initio model EFT by default.



d+Au multiplicity used in centrality cut vs. $\langle N_{part} \rangle$ and $\langle N_{coll} \rangle$

Similar using Nch and total MIP. Use Nch for $\langle N_{part} \rangle$ and $\langle N_{coll} \rangle$. Or, if analysis require centrality based on the multiplicity from forward rapidity, use total MIP.



Plots and tables

Every model (in O+O) and multiplicity has associated tables and plots detailing their <Npart> and <Npart> cuts, systematic and statistical uncertainties.

Please see the Big Summary: https://drupal.star.bnl.gov/STAR/system/files/analysis_summary.pdf

Summary

1. NBD fit is equally good for different models and multiplicities, and $\langle N_{\text{part}} \rangle$, $\langle N_{\text{coll}} \rangle$ show little change with respect to models and multiplicities.
2. **For O+O use the Nch and EFT** model based values as the default for centrality cuts or $\langle N_{\text{part}} \rangle$, $\langle N_{\text{coll}} \rangle$, unless analysis requires special forward rapidity multiplicity based centrality or requires special models.
3. **For d+Au use Nch**, unless analysis requires special forward rapidity multiplicity based centrality.
4. Statistical unc. on the $\langle N_{\text{part}} \rangle$, $\langle N_{\text{coll}} \rangle$ is negligible. It may not be needed. Only quote systematic errors.
5. Float centrality cuts were shown for more detail, but STAR standard is to use integer and only in 5% increments up to 80%, and those integers are simply rounded closest integers to the float numbers. StRefMultCorr will use those integers. If finer bins are used, they are not centrality cuts, but simply finer bins in that specific multiplicity. For example, top 1% Nch($|\eta| < 1.5$) events, not 1% centrality events. This is because resolution of small centrality interval is hard when multiplicity smearing is relatively large compared to multiplicity values in those systems.